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RESEARCH ARTICLE

Unmasking the Fake: Machine Learning Approach for Deepfake Voice Detection

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ABSTRACT Deepfake voice refers to artificially generated or manipulated audio that mimics a person's voice, often created using advanced AI techniques. These synthetic voices can be used to convincingly imitate someone, making them nearly indistinguishable from genuine recordings. We present an advanced method for deepfake voice detection, leveraging a custom model named MFCC-GNB XtractNet. By extracting Mel-Frequency Cepstral Coefficients (MFCC) from audio samples which serve as the foundational features for identifying genuine and fake voices. These MFCC features are then enhanced through a transformation process that employs a Gaussian Naive Bayes (GNB) model in conjunction with Non-Negative Factorization, creating a more discriminative feature set for subsequent analysis. These features are fed to our developed model, MFCC-GNB XtractNet to identify deep fake voice.To rigorously evaluate the effectiveness of our approach, we deployed a range of machine learning models, including Random Forest (RF), K-Nearest Neighbors Classifier (KNC), Logistic Regression (LR) and Gaussian Naive Bayes (GNB). Each model's performance is assessed through k-fold cross-validation, ensuring a robust evaluation across multiple data splits. Additionally, we performed a computational cost analysis to measure the efficiency of the models in terms of training time and resource usage. The results of our experiments were highly promising, with our MFCC-GNB XtractNet + GNB model achieving an impressive accuracy score of 99.93%. This exceptional performance underscores the model's ability to effectively distinguish between real and deepfake voices setting a new benchmark in the field of voice authentication.

INDEX TERMS Deep fake voice, machine learning, MFCC-GNB XtractNet, transfer learning.

I. INTRODUCTION

Deep fake voice technology refers [1] to the use of artificial intelligence (AI) to create synthetic audio that closely mimics a human voice. By training on large datasets of voice recordings deep learning models can generate speech that sounds [2] like a specific person, often with startling accuracy. This technology can be used for various purposes, such as creating realistic voiceovers, dubbing, or even mimicking someone's voice for entertainment or creative projects. However, it also raises ethical concerns particularly in relation to privacy, consent, and the potential for misuse in activities like impersonation or spreading misinformation.

Deepfake voice technology poses significant [3] dangers due to its ability to convincingly imitate the voices of real people. One of the primary risks is identity theft, where malicious actors can use a cloned voice to deceive others into divulging sensitive information or authorizing transactions. Additionally, deepfake voices [4] can be used to spread misinformation, such as creating fake audio recordings that appear to be from trusted public figures, leading to public panic or manipulation of public opinion. This technology can also be exploited for personal attacks, such as creating fake evidence in legal disputes or harassing individuals by impersonating their voice. The potential for deepfake voice technology to be used in fraudulent or harmful ways highlights the urgent need for robust detection methods and legal frameworks to protect individuals and society from its

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misuse. As deepfake voice technology continues to evolve it underscores the need for clear [5] guidelines and safeguards to prevent its exploitation.

Advanced machine learning and transfer learning [6] are crucial for detecting deepfake voices due to the increasingly sophisticated nature of these synthetic audio forgeries. As deepfake technology evolves, it becomes harder to distinguish between [7] genuine and fake voices using traditional methods. Advanced machine learning models can analyze subtle patterns and inconsistencies in audio data that may be missed by the human ear or simpler algorithms. Transfer learning [8] in particular, is essential because it allows models to leverage knowledge gained from large datasets and apply it to new, smaller datasets, improving detection accuracy even when examples of deepfake voices are limited. By continually refining and updating these models, we can [9] stay ahead of the rapidly advancing deepfake technology and protect against its misuse in impersonation, fraud, and misinformation.

In this study, we focus on detecting deepfake voices by employing a combination of machine learning and transfer learning techniques. Our process begins by extracting Mel-Frequency Cepstral Coefficients (MFCC) from the audio data, which serve as the initial set of features for training various machine learning models. These models are then used to analyze the MFCC features and identify potential deepfakes.To enhance our detection capabilities, we introduce a transfer learning step. Here, we use a Gaussian Naive Bayes (GNB) model to generate new probabilistic features, which are further refined using non-negative matrix factorization (NMF). These transformed features provide a different perspective on the data, capturing patterns that might be overlooked in the original feature set. We apply the same machine learning models to this new set of features to evaluate how well they improve deepfake voice detection. By comparing the performance of the models on both the original MFCC features and the newly derived probabilistic features, we aim to determine the effectiveness of this combined approach in accurately identifying deepfake voices.

Our main research contributions are as follows:

- A key novelty of our approach lies in the introduction of MFCC-GNB XtractNet a hybrid model that uniquely combines MFCC feature extraction with Gaussian Naive Bayes (GNB) and non-negative factorization. This integration allows for enhanced feature transformation and detection accuracy in identifying deepfake voice recordings, setting it apart from conventional deepfake detection methods.
- The novelty of our approach, MFCC-GNB XtractNet, extends beyond its unique combination of MFCC feature extraction and GNB-based transformation. We applied k-fold validation analysis to rigorously evaluate the model's performance, ensuring robust and reliable results. Additionally, we assessed the computational cost of the deployed model, providing insights into its efficiency and practicality.

II. BACKGROUNDS

The current review has aimed at trying to review and organize the findings in an effort to determine the existing gaps, trends as well as emerging themes that form the basis of the current study. Table 1 presents a summary of the literature studied.

A. PREDICTIVE ANALYSIS WITH MACHINE LEARNING

The research study in [10] delves into how our innate biological processes can help us distinguish between real and cloned human voices. The idea is that certain natural features in speech, like pauses, can be strong indicators of authenticity. To explore this, the researchers gathered voice samples from 49 adults with diverse ethnic backgrounds and accents. These samples were used to train three different voice cloning models and create control paragraphs for comparison. After testing five machine learning models, the AdaBoost model stood out achieving accuracy of 0.79 when predicting unseen data. This research suggests that by paying attention to subtle cues in speech, we can improve our ability to detect cloned voices, offering a promising direction for future developments in this area.

Hamza et al. has focused on using machine and deep learning techniques in [11] to detect deepfake audio, with Mel-frequency cepstral coefficients (MFCCs) being key to extracting important features from the audio. The study uses the Fake-or-Real dataset, a recent benchmark that includes four sub-datasets—For-rece, For-2-sec, For-norm, and For-original—each categorized by audio length and bit rate. The findings show that the Support Vector Machine (SVM) model performed best on the For-rece and For-2-sec datasets, while the Gradient Boosting model excelled with the For-norm dataset. The VGG-16 model also stood out, delivering impressive results on the For-original dataset and surpassing other advanced methods. The SVM model, in particular, achieved a remarkable accuracy of 98.83%, underscoring its strong capability in detecting deepfake audio.

The increased use of audio in various aspects of life, audio forgery detection has thus been recognized as very vital aspect of forensic analysis, this is so since the authenticity of digital evidence is very key when it comes to the court of law. Due to the advancement in technology and solving puzzles, advanced methods of faking content like imitation whereby a speaker closely imitates another person's voice through signal processing then one must have reliable methods of detecting the same. The work in [12] consolidates a passive forgery detection strategy without necessarily requiring the user's interference; instead, the user extracts the entropy features from both the genuine and the imitated audio contents. These features are then classified by using the method of logistic regression model. The approach appeared to be very successful with the accuracy of 98% all of the forged audios were caught.

[13] specifically deals with the relatively newer Fake-or-Real (FoR) dataset which, contains audio samples synthesized from TTS systems. The paper outlines strategies that focus on the best feature extraction and the identification of the most suitable machine learning algorithms to distinguish between fake and real audio. Feature engineering means extracting different aspects of audio, while the feature selection techniques help to find the best minimal set of features with the highest accuracy of the classifiers, which is then used as inputs. Six machine learning classifiers are compared in three subsets of the FoR dataset within the research framework of this study. This proves that the proposed method in the work enhances the identification of fake audios as compared to the baseline method with a 26% increase in the accuracy score. It is noteworthy that the highest accuracy among the models was obtained while using the XGBoost (XGB) model 93%.

The scale and scope of voice disguise remain [14] a major concern in FASR and multimedia applications where speaker identification is accurate. In the art of voicing criminals are able to employ various techniques the disguise techniques consist of pitch modification of the raised, lower, fast, and slow pitch disguises which exhibit different acoustic properties concerning mean and correlation coefficients. It focuses on examining the effectiveness of classification for identifying masked voices in relation to gender differentiation of the samples utilising new achievements in the field of acoustic features. Normal and four forms of disguised voices were analyzed in order to extract MFCCC features. For speaker classification, feature-based classifiers k-NN and support vector machine were selected. These findings show that SVM classifiers are more effective than k-NN: normal voices have a classification efficiency of 98.58% for both classifiers. These results highlight that SVMs along with MFCC based features plays a significant role in resolving the challenges of disguised voice analysis in both forensic and multimedia domain.

B. PREDICTIVE ANALYSIS WITH DEEP LEARNING

With the emergence of new improved software technologies of its generation and manipulation the number of cases that cover the generation or manipulation of digital voice has increased considerably and is a clear indication that the malefactors are showing a keen interest in the generation of digital voice and manipulation and since the technology out there has equally advanced so much, hence the call for efficient methods of detection of the authenticity of the voice was imperative. In [15], it has been found that a combination of convolutional neural networks (CNN) and Mel spectrograms seems to be rather effective in detecting fake voices. Besides, a supervised experimental evaluation was administered with speech samples from several voice datasets so as to compare this CNN for this detection task across spoken languages. The accuracy was improved at a percentage of 99% for FoR, 94% for ASV, and 98% for WaveFake dataset when trained and tested on a single set of datasets respectively; and 98% for FoR, 92% for ASV, and 96% for the WaveFake dataset when the model was trained on a merged dataset but tested on each of the datasets. As a result, the study complies with the indexes of the model at the state of the art, which indicates the relevance and efficiency of employing the model for preventing the malicious use of synthetic voices.

The research in [16], While it examines the integration of XAI into deepfake voice detection in order to create an additional human perception level. Due to this, the research sets its goals at reaching non-experts in both linguistics and artificial intelligence making it more focused on ease of use instead of speed. Although, models such as transformers, conformers and wav2vec are some of the most used for the task of speech recognition and speaker verification due to the models' high performance, this paper uses more basic models intentionally. Feature extraction was done from spectrograms which though have relatively lower preprocessing demands but have higher interpretability. This work's primary contributions are twofold: Firstly, transferring the XAI methodology to the LSTM domain and, secondly, innovating with a multi-modal interpretability approach. While the focus is on simplicity, the approached proposed in the paper obtained a performance score if 98%, which is indicative of the framework's utility in deepfake detection while remaining easy to use.

Revealed how the adversaries have used fake audio for [17] several malicious intentions, thus creating high challenges when investigating the authenticity of audio during forensic analysis. This review is specifically about the deepfake audio classification and aims at identifying the problems involved and appraising the existing detection approaches within the forensic environment. Audio analysis was done with features such as MFCC, Mel-spectrum, Chromagram, and spectrograms in a bid to observe characteristics that distinguished the sounds. In this context, the study finally compares the various deep learning approaches identified from previous work in five rounds of experiments that measure their mean accuracy as well as impacts. Preliminary findings showed that Chromagram, Spectrogram, and Mel-spectrum produced a better outcome when a custom architecture was used to create the model while the MFCC frames were better aligned to VGG-16 architecture that scored 88%.

The phenomenon of malicious speech which on the surface may be perceived as originating from a person, artificial intelligence, or signal processing technology, is a major issue in the modern world. Techniques such as Deep Voice which produces somewhat synthesized sounds is used to create fake audio whereas Imitation which give more realistic sounds is used to create fake audio. Identifying such fake content is not very easy, and this is due to the large number of voice recordings that is shared online. To tackle this, the formulation of a CNN with image augmentation and dropout was presented as a solution in [18]. The model was trained on 2,092 histograms of genuine and fake voice samples and

cross-checked with 864 histograms which impressively give a global accuracy of 98%.Therefore, these results show that the system can be efficient in establishing the presence of fake voice content.

The research work in [19] has focused on issues of synthesizing speech and identifying speakers according to specific parameters with the help of deep learning in the framework of text-to-speech (TTS) systems. A new method is introduced to detect speech fakes in real multi-person interactions. It takes into account multiple Deep Neural Network models and several datasets both for training and testing such as Urban-Sound8K, Conversational, AMI-Corpus, and FakeOrReal. The approach comprises four primary components: A pre-processing module, the speech-denoising module that takes audio feed and uses Multiple Layer Perceptron and convolutional neural network for denoising of the audio with accuracies of 93% and 94% respectively; a text conversion using Natural Language Processing with 93% efficiency and speaker labelling using Recurrent Neural Network with the efficiency of 80% and Diarization Error Rate of 0.52 and the last one with the use of CNN that targets to differentiate between the original and fake audio with a precision of 94%.

C. RESEARCH GAPS AND LIMITATIONS

After reviewing previous research, we discovered some gaps.

- A notable research gap addressed by our work is the reliance on classical approaches in previous studies, which often lacked advanced mechanisms for deepfake voice detection. While traditional methods have been used, they are limited by their lower accuracy and inadequate handling of the complexities of synthetic speech. Furthermore, earlier deep learning models, despite their potential, suffered from high computational costs and suboptimal accuracy scores. Our approach, MFCC-GNB XtractNet, aims to bridge this gap by leveraging advanced feature extraction and transformation techniques, improving detection accuracy while managing computational efficiency.

III. PROPOSED METHODOLOGY

Our methodology begins with data preprocessing on deepfake voice recordings to ensure the data is clean and suitable for analysis. The Figure 1 shows our innovative methodology approach. After preprocessing and balancing the dataset we extract Mel-Frequency Cepstral Coefficients (MFCC) features from the audio data, which serve as the foundation for subsequent model training. We then apply machine learning models to these MFCC features to establish baseline performance. Next, we enhance the feature set by transferring new features using a combination of Gaussian Naive Bayes (GNB) and non-negative factorization techniques. This transformed feature set is then used to train and evaluate additional machine learning models.This approach allows us to leverage both traditional feature extraction and advanced feature transformation methods to improve deepfake voice detection accuracy.Our approach goes beyond the direct use of MFCC features by incorporating a transfer learning step. Specifically, we transform the MFCC-derived features into a new probabilistic feature space using a GNB model, followed by further refinement through NMF. This step enables the model to capture additional patterns and relationships within the data that might not be directly apparent in the original MFCC features.

A. DEEPFAKE VOICE DATASET

For our research, we used a publicly [20] available deepfake voice dataset. The dataset's raw audio files are organized in the "AUDIO" directory, divided into "REAL" and "FAKE" class directories. Each filename specifies the original speaker and the transformed voice. Additionally, we utilized the "DATASET-balanced.csv" file for feature extraction. This file contains features derived from one-second audio segments, with data balanced through random sampling to ensure an equitable distribution of real and fake audio samples.

B. DATA PREPROCESSING

Voice data preprocessing for deepfake detection involves several critical steps to ensure the quality and relevance of the audio data. First, raw audio files are cleaned to remove noise and normalize volume levels, which helps in minimizing the impact of recording inconsistencies. Next, audio files are segmented into manageable chunks, typically one-second windows, to facilitate feature extraction. During this segmentation, any irrelevant or corrupted segments are filtered out. Subsequently, the audio data is converted into a suitable format for feature extraction. This transformation captures essential characteristics of the audio signal while reducing dimensionality. Finally, the preprocessed audio is organized into categories—real and fake—based on its source to prepare it for further analysis and model training. This comprehensive preprocessing pipeline ensures that the data is clean, consistent, and ready for effective deepfake voice detection.

C. SMOTE-BASED BALANCING

To address the class imbalance in deepfake voice data [21], we applied SMOTE-based resampling. SMOTE, or Synthetic Minority Over-sampling Technique, is a powerful method for generating synthetic samples of the minority class by interpolating between existing data points. By applying SMOTE, we effectively increased the representation of the minority class in our dataset, which helps to balance the distribution of real and fake voice samples. Figure 2 shows the plots after applying smote. This technique enhances the model's ability to learn from the minority class and improves overall performance by providing a more even distribution of examples for training, thereby addressing common challenges associated with imbalanced datasets in deepfake voice detection.

TABLE 1. The literature analysis based on state-of-the-art approaches performance.

Ref	Year	Purposed Technique	Dataset	Performance Score	Research Gap
[10]	2024	AdaBoost Classifier	Speech recogniiton Dataset	79.6%	Classical approach is used and accuracy score is low.
[11]	2022	Support vector machine	Four sub-datasets—For-rece, For-2-sec, For-norm, and For-original	98.3%	Classical approach is used.
[12]	2020	Logistic regression	Audio forgery dataset	98.0%	Traditional approach is used and Accuracy score is low.
[13]	2022	XGBoost classifier	Real and fake audio sample Dataset	93.0%	Performance score is low.
[15]	2024	Deeplearning based CNN framework	Real and fake voice dataset	93.0%	Computational cost is high.
[16]	2022	LSTM Model	Fake voice dataset	98.0%	Dataset size is small and computational cost is increased.
[17]	2023	Deep learning based VGG-16 framework	Fake voice classification dataset	99.0%	Computational cost is increased.
[19]	2020	CNN framework	Fake and real voice dataset	94.0%	Dataest size is small and computational cost is high.
[18]	2021	CNN model	Fake voice content dataset	94.0%	Generalizability of Dataset and computational cost is high.

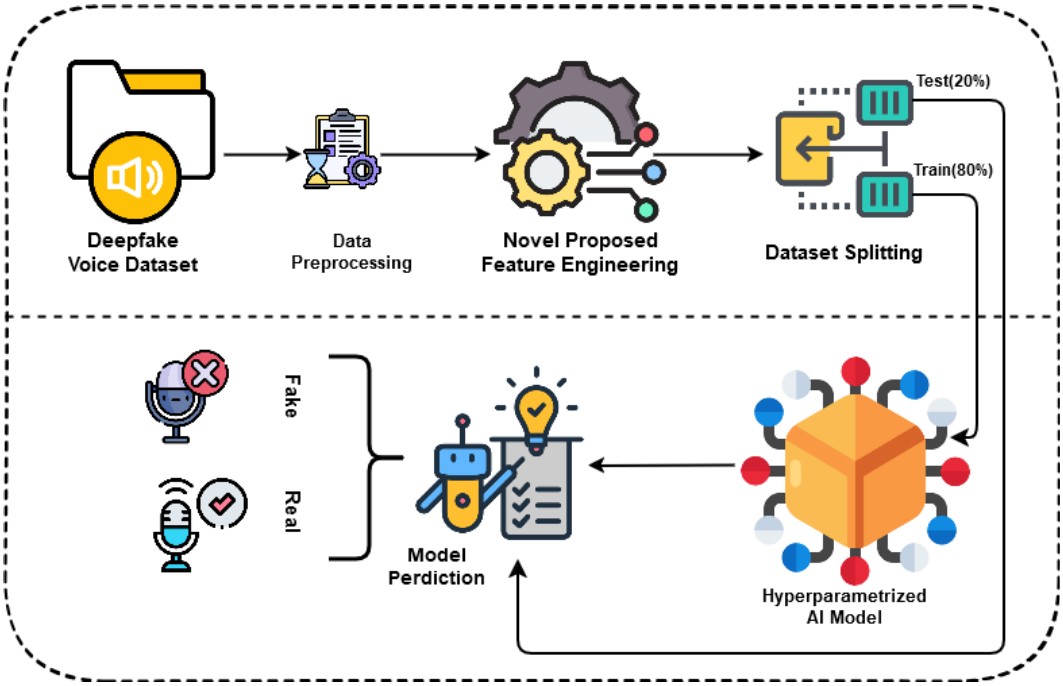

FIGURE 1. Our innovative approach methodology for the deepfake voice detection.

D. MFCC FEATURES

To enhance the detection of deepfake voices [14] we extract Mel-Frequency Cepstral Coefficients (MFCC) from the audio recordings. MFCCs are widely used in speech and audio processing because they capture the essential spectral features of an audio signal in a manner that closely aligns with human auditory perception. The extraction process involves converting raw audio data into a series of coefficients that represent the power spectrum of short, overlapping windows of audio. This transformation effectively reduces the dimensionality of the data while preserving key acoustic properties, making it easier to differentiate between genuine and synthetic voices. By focusing on these coefficients, our approach leverages their sensitivity to subtle variations in vocal characteristics, which is critical for distinguishing deepfake voices from real ones. This feature extraction technique improves the model's ability to detect even subtle manipulations in voice recordings, thereby enhancing overall detection accuracy and reliability.

E. PROPOSED MFCC-GNB XTRACTNET APPROACH

Our methodology begins with the extraction of Mel-Frequency Cepstral Coefficients (MFCC) from the audio data, which provides a detailed representation of the spectral

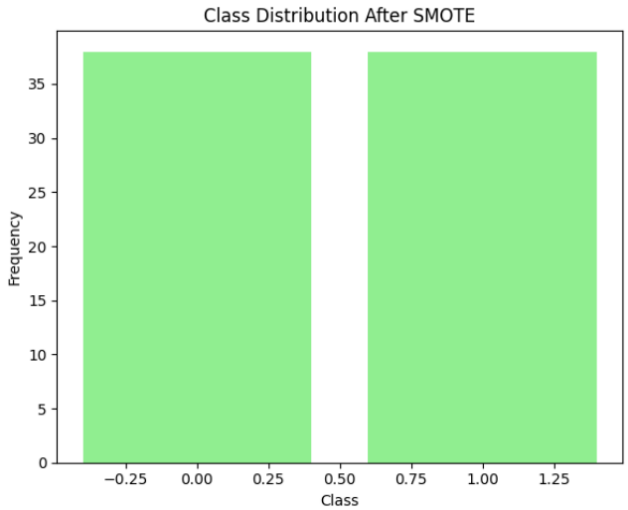

FIGURE 2. After applying smote for balance the dataset.

characteristics of the voice recordings. MFCCs are particularly effective for capturing the nuances of audio signals making them ideal for voice analysis. Figure 3 shows our innovative feature engineering process. Following this, we employ a two-tiered transformation process to enhance the extracted features. First, we use a GNB model to refine and classify the MFCC features leveraging the model's probabilistic approach to improve feature distinction. Subsequently, we apply Non-negative Factorization to further decompose and transform these features, ensuring that the resulting feature set is both informative and representative of the underlying audio patterns. This combined approach, known as MFCC-GNB XtractNet, integrates sophisticated techniques to optimize feature extraction and transformation, thereby enhancing the model's capacity to accurately detect deepfake voices. By utilizing MFCCs in conjunction with GNB and Non-negative Factorization, our method addresses complex variations in voice recordings and improves overall detection performance. NMF has demonstrated significant performance improvements when combined with MFCC for feature extraction particularly in audio processing tasks. MFCC captures critical spectral features by representing audio signals in a way that aligns closely with human auditory perception, making it a widely used technique in voice and speech analysis. However, raw MFCC features may lack refined interpretability or discriminative power when directly used for classification. NMF addresses this limitation by decomposing the MFCC feature matrix into two non-negative matrices: a basis matrix and a coefficient matrix. This decomposition enhances feature representation by uncovering latent patterns and suppressing noise, effectively isolating the most informative components of the MFCC features. As a result, the combination of NMF with MFCC not only preserves the essential characteristics of the audio signal but also enhances the discriminative capability of the features.

Algorithm 1 Deepfake Voice Detection: MFCC-GNB XtractNet Process

Audio recordings A_i, Gaussian Naive Bayes model f_{GNB}, Non-negative Matrix Factorization f_{NMF}, Machine learning classifier f_{ML} Predicted label $\hat{y}_{\text{audio}}$ for audio recordings A_{audio}

Step 1: Data Preprocessing Preprocess input audio to ensure clean data:

$$A_i' \leftarrow \text{preprocess}(A_i)$$

Balance the dataset if necessary:

$$A_{\text{balanced}} \leftarrow \text{balance}(A_i')$$

Step 2: Feature Extraction Extract MFCC features from the balanced audio:

$$\mathbf{f}_{\text{MFCC}} \leftarrow f_{\text{MFCC}}(A_{\text{balanced}})$$

Step 3: Feature Enhancement with GNB Refine and classify features using Gaussian Naive Bayes:

$$\mathbf{f}_{\text{GNB}} \leftarrow f_{\text{GNB}}(\mathbf{f}_{\text{MFCC}})$$

Step 4: Feature Transformation with NMF Enhance features using Non-negative Matrix Factorization:

$$\mathbf{f}_{\text{NMF}} \leftarrow f_{\text{NMF}}(\mathbf{f}_{\text{GNB}})$$

Step 5: Classification Train machine learning classifier on transformed features:

$$\hat{y}_i \leftarrow f_{\text{ML}}(\mathbf{f}_{\text{NMF}})$$

Step 6: Evaluation Compute evaluation metrics:

$$\text{Metrics} \leftarrow \{\text{Accuracy, Precision, Recall, F1-Score, AUC}\}$$

Step 7: Prediction for Audio Recordings For new audio recordings A_{audio}, predict:

$$\hat{y}_{\text{audio}} \leftarrow f_{\text{ML}}(f_{\text{NMF}}(f_{\text{GNB}}(f_{\text{MFCC}}(A_{\text{audio}}))))$$

F. MACHINE LEARNING APPROACHES

We applied four advance Ml models including; GNB.LR,RF and KNC model for the real time IoT attack detection. Basic workign and characteristic of these models are discussed below.

G. GAUSSIAN NAIVE BAYES

In the context of real and fake voice detection [22] the Gaussian Naive Bayes (GNB) model is employed to classify audio features based on their probabilistic distribution. The GNB model assumes that the features follow a Gaussian distribution within each class, simplifying the classification problem by applying Bayes' theorem with a strong independence assumption between features. For a given audio sample x, the model computes the posterior probability of the sample belonging to a class y

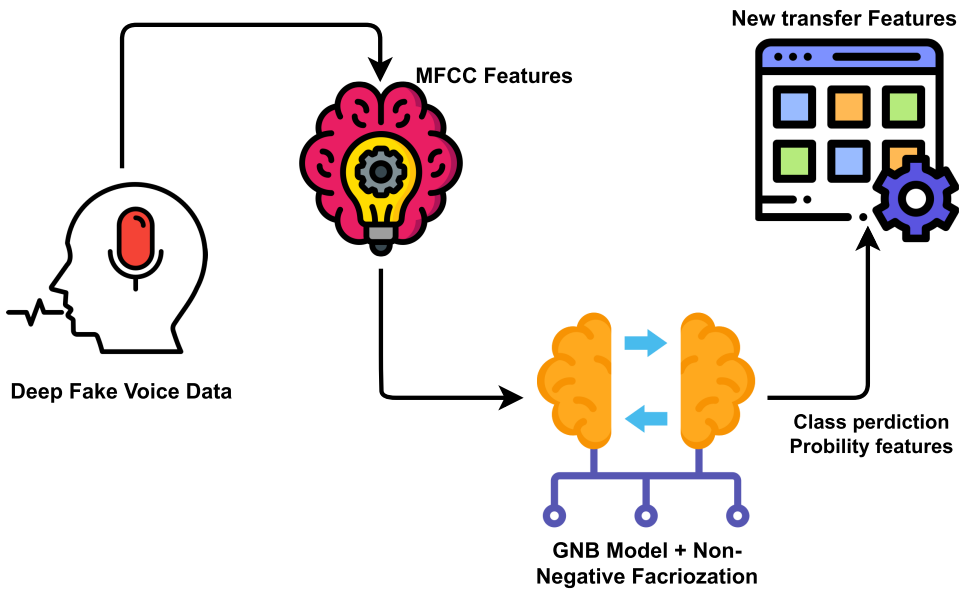

FIGURE 3. Our proposed architectural analysis of the workflow for our innovative feature engineering process for the deep fake voice detection.

as follows:

$$P(y \mid x) = \frac{P(x \mid y) \cdot P(y)}{P(x)} \tag{1}$$

where $P(y \mid x)$ is the posterior probability of class y given the feature vector x, $P(x \mid y)$ is the likelihood of x given class y, $P(y)$ is the prior probability of class y, and $P(x)$ is the probability of the feature vector x. In practice, $P(x \mid y)$ is modeled as a Gaussian distribution:

$$P(x_i \mid y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right) \tag{2}$$

where μ and σ^2 are the mean and variance of the feature x_i for class y. This approach allows the GNB model to effectively classify audio samples as real or fake based on the learned distributions of the features providing a robust method for detecting deepfake voices.

H. LOGISTIC REGRESSION

Logistic Regression (LR) is a widely used method [23] for binary classification tasks, including deepfake voice detection. In this context, LR models the probability of an audio sample belonging to a particular class—real or fake—based on its features. The model uses the logistic function to output probabilities, which are then mapped to class labels. For a given feature vector $\mathbf{x}$, the probability $P(y = 1 \mid \mathbf{x})$ that the sample belongs to the positive class (e.g., fake voice) is computed as follows:

$$P(y = 1 \mid \mathbf{x}) = \frac{1}{1 + \exp(-(\mathbf{w}^\top \mathbf{x} + b))} \tag{3}$$

where $\mathbf{w}$ represents the vector of model coefficients, b is the bias term, and $\mathbf{x}$ is the feature vector. The function $\mathbf{w}^\top \mathbf{x} + b$ is the linear combination of the features, which is passed through the sigmoid function $\frac{1}{1+\exp(-z)}$ to obtain a probability score between 0 and 1. This probability is then used to classify the audio sample as either real or fake. Logistic Regression's simplicity and interpretability make it a valuable tool for detecting synthetic audio content, as it effectively models the relationship between audio features and class probabilities.

I. RANDOM FOREST

Random Forest (RF) is a robust ensemble learning technique [24] that is effective for binary classification tasks such as deepfake voice detection. It operates by constructing multiple decision trees during training and outputting the mode of the classes (classification) or mean prediction (regression) of the individual trees. Each decision tree is built using a random subset of the features and training samples, which helps in reducing overfitting and improving model generalization. For a given audio sample $\mathbf{x}$, the Random Forest model aggregates the predictions of N individual decision trees to make the final classification. Mathematically, the prediction $\hat{y}$ of the Random Forest expressed as:

$$\hat{y} = \text{mode}\left(\{T_1(\mathbf{x}), T_2(\mathbf{x}), \ldots, T_N(\mathbf{x})\}\right) \tag{4}$$

where $T_i(\mathbf{x})$ is the prediction of the i-th decision tree, and N is the total number of trees in the forest. Each tree T_i outputs a class label for the input feature vector $\mathbf{x}$, and the mode function selects the most frequent class label among all trees. This ensemble approach leverages the diversity among trees to enhance classification accuracy and stability, making it a powerful tool for distinguishing between real and fake audio samples in deepfake voice detection.

J. K-NEIGHBOUR CLASSIFIER

K-Nearest Neighbors (KNN) is an intuitive and straightforward [25] machine learning algorithm widely used for classification tasks, such as detecting deepfake voices. KNN classifies a new audio sample by examining its K closest neighbors in the feature space and assigning the most common class label among these neighbors. The algorithm operates on the assumption that similar data points tend to share the same class label. For an audio sample $\mathbf{x}$, the KNN algorithm finds the K training samples that are closest to $\mathbf{x}$ based on a chosen distance metric, such as Euclidean distance, and assigns the label that is most frequent among these nearest neighbors.

Mathematically, the classification $\hat{y}$ for the sample $\mathbf{x}$ is given by:

$$\hat{y} = \text{mode}\left(\{y_i \mid \mathbf{x}_i \text{ is among the } K \text{ nearest neighbors of } \mathbf{x}\}\right) \tag{5}$$

Here, y_i represents the class label of the i-th nearest neighbor, and the function mode determines the most common label among these neighbors. By relying on local feature similarity, KNN can effectively identify whether an audio sample is real or fake, making it a valuable tool for deepfake voice detection.

K. HYPERPARAMETER TUNING

The selection of the optimal hyperparameters [26] for the machine learning algorithms is summarized in the table 2 below. Each model is fine-tuned using k-fold cross-validation, with daily backups and rigorous testing and training procedures. Our research indicates that this meticulous hyperparameter tuning process is essential for achieving the highest performance scores in detecting deepfake images.

TABLE 2. Selected values of hyperparameters of applied models. This study shows hyperparameter values of applied models.

Technique	Hyperparameter
LR	$\text{copy}_x = ' \; True fitintercept = ' \; True', Positive = ' False', Normalize = ' False', iteration = 300$
RF	max-depth=10,nestimators=50,criterion='gini', $\text{num}_l eaves = 31, learning_r ate = 0.1.$
GNB	Var-Smoothing,1e-9,estimator=50,learning-rate=0.1.
KNC	Laplace Smoothing= 1,Variance Smoothing=0.

L. PRIVACY PROTECTION

Privacy Protection Ensuring that the audio data used for training and detection is anonymized and securely stored to prevent misuse or unauthorized access. Responsible Deployment Advocating for the use of this technology strictly in scenarios where it upholds security and privacy such as protecting individuals and organizations from fraud while discouraging applications that could infringe on personal rights. Bias and Fairness Highlighting the importance of using diverse datasets to ensure the model performs equitably across different languages, dialects and demographics to avoid potential biases. Transparency Encouraging transparency in the deployment of deepfake detection systems, including clear communication with stakeholders about the purpose, capabilities and limitations of the technology. Ethical Guidelines Suggesting that organizations adopting this technology adhere to established ethical frameworks such as the principles of fairness, accountability and transparency in AI. Source code is an available online repository [27] for the development of research community and collaborations.

IV. RESULTS AND DISCUSSIONS

In this section, we provide breifly discussion of the results achieved with our innovative approach. We highlight key findings and insights derived from the application of our method, focusing on its effectiveness and impact.

A. EXPERIMENTAL SETUP

In our experiment, we used Google Colab, a cloud-based platform that runs Jupyter Notebooks, to execute the analysis. The specifications of the setup are detailed in the table provided. The performance of our machine learning approach was evaluated using multiple metrics, including accuracy, F1 score, precision, and recall.

TABLE 3. The experimental setup information.

Specification	Value
Model	Dell Intel(R) CPU 3.00Ghz
ProgrammingLanguage	Python 3.2
CPUMHZ	3000
Cachevolume	5632KB
RAM	16 GB
CPUcore	1
AddressVolume	32bits Virtual , 48 bits Physical

B. RESULTS WITH MFCC FEATURES

In our analysis, we evaluated several machine learning models on MFCC features to determine their effectiveness in deepfake voice detection. The models tested were Gaussian Naive Bayes (GNB), Logistic Regression (LR), Random Forest (RF), and K-Nearest Neighbors (KNC).The table 4 indicates classification report. The Random Forest model achieved a performance score of 90%, indicating its high efficacy in handling the complexity of the data. The K-Nearest Neighbors model followed with a score of 84%, demonstrating strong performance but slightly less effective compared to Random Forest. Logistic Regression recorded a performance score of 75%, showing a more modest effectiveness in comparison. The Gaussian Naive Bayes model excelled with a score of 92%, highlighting its superior performance in classifying MFCC features. These results underscore the relative strengths of each model, with GNB and RF performing notably well in distinguishing between real and fake voices.

TABLE 4. Performance analysis with MFCC features and new transfer features.

Models	Accuracy with (MFCC)	Accuracy with New Features	Target Class	Precision	Recall	F1 Score
RF	0.90	0.92	Fake	0.92	0.94	0.96
			Real	0.98	0.50	0.67
			Average	0.90	0.94	0.96
LR	0.75	0.84	Fake	0.87	0.91	0.87
			Real	0.00	0.00	0.01
			Average	0.75	0.75	0.75
GNB	0.92	0.9993	Fake	0.92	1.00	0.96
			Real	1.00	0.50	0.67
			Average	0.92	0.92	0.92
KNC	0.84	0.93	Fake	0.85	0.96	0.92
			Real	0.00	0.00	0.00
			Average	0.84	0.84	0.84

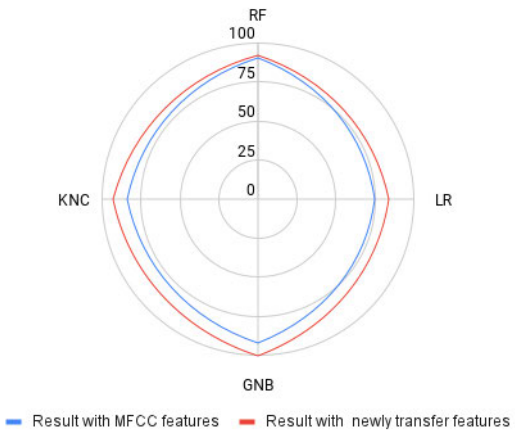

FIGURE 4. Our proposed approach radar chart analysis.

C. RESULTS WITH OUR PROPOSED APPROACH

We reassessed the performance of various machine learning models using the newly transformed features from our proposed approach. The table 4 indicates the complete classification report. The results revealed notable improvements in classification accuracy compared to previous evaluations. Specifically, Random Forest achieved an impressive score of 92.30%, demonstrating its continued effectiveness with the new features. K-Nearest Neighbors performed slightly better, reaching 93%, indicating a robust ability to handle the updated data. Logistic Regression showed a moderate score of 84%, reflecting its stable performance with the new feature set. The Gaussian Naive Bayes model delivered outstanding results with a near-perfect accuracy of 99.93%, highlighting its exceptional capability to distinguish between real and fake voices using enhanced features. The figure 4 shows radar chart-based analysis with mfcc features and our newly transferred features. These findings underscore the efficacy of our feature transformation approach, which significantly enhanced the performance of all evaluated models, particularly boosting the accuracy of Gaussian Naive Bayes to near-optimal levels.

TABLE 5. Our novel proposed shows state-of-the-art k-fold validation analysis.

Technique	Kfold	K-Fod Accuracy (%)	Standard Deviation (%)
KNC	10	0.96	+/-0.9433
LR	10	0.8867	+/-0.1492
Rf	10	0.96	+/-0.0800
GNB	10	0.96	+/-0.0800

D. KFOLD VALIDATIONS ANALYSIS

We applied a 10-fold cross-validation analysis to evaluate the performance of several models, achieving the following results: the Random Forest (RF) model attained an accuracy of 0.9600 with a standard deviation of 0.0800, indicating consistent performance across folds. The K-Nearest Neighbors Classifier (KNC) achieved an accuracy of 0.9433 with a standard deviation of 0.0870, also demonstrating reliable results. The Logistic Regression (LR) model yielded an accuracy of 0.8867 with a higher standard deviation of 0.1492, suggesting some variability in its performance. Lastly, the Gaussian Naive Bayes (GNB) model matched the RF's performance with an accuracy of 0.9600 and a standard deviation of 0.0800.

E. COMPUTATIONAL COMPLEXITY ANALYSIS

In the computational complexity analysis, we measured the training times of various models.The table 6 shows computational complexity analysis. The Random Forest (RF) model required the most time, with a training duration of 0.0180411 seconds reflecting its higher computational demands due to the ensemble of decision trees. The K-Nearest Neighbors Classifier (KNC) followed with a training time of 0.0078217 seconds indicating moderate computational complexity. The Logistic Regression (LR) model

TABLE 6. Shows computational complexity analysis.

Technique	Computation Time
RF	0.0180411s
LR	0.0060997s
GNB	0.00576972s
KNC	0.0078217s

was slightly faster, with a training time of 0.0060997 seconds, highlighting its relatively efficient training process. The Gaussian Naive Bayes (GNB) model had the shortest training time at 0.00576972 seconds, demonstrating its computational efficiency and suitability for quick model training.The scalability of our MFCC-GNB XtractNet model is a key strength, allowing it to efficiently handle large datasets and adapt to diverse real-world applications. The model's design ensures that as the volume of input data increases the processing time and resource requirements remain manageable. This scalability is achieved through optimized feature extraction using MFCC which captures the most essential characteristics of the audio data without excessive computational burden. Additionally, the two-tiered feature transformation process involving GNB and Non-negative Factorization is computationally efficient, allowing the model to scale across varying dataset sizes. This makes the approach suitable for deployment in systems that must process large-scale data streams such as real-time voice analysis platforms for deepfake detection, ensuring robust performance even in high-demand scenarios.

F. STATE OF THE ART STUDIES COMPARISONS

We conducted a comprehensive performance comparison between our proposed solution and existing approaches documented in the literature. The previous research frequently employed various deep learning and machine learning methods. In contrast, our study introduced an advanced feature transfer mechanism with an innovative strategy, significantly enhancing performance. While the highest accuracy achieved by past approaches was 99%, our proposed technique demonstrated superior effectiveness achieving a remarkable performance score of 99.93%. The figure 5 indicates confusion matrix heatmap graph analysis. This analysis indicates that our innovative approach outperforms previous methods setting a new benchmark in the field.

TABLE 7. Our novel proposed shows the comparison with the state of the art past applied approaches.

Ref	Year	Purposed Technique	Dataset	Accuracy Percentage
[10]	2024	Adaboost classifier	speech recognition voice dataset	79.0
[15]	2024	CNN model	Real and fake voice dataset	93.0
[28]	2024	Deep Ensemble model	ASV Spoof2019 dataset	99.0
[17]	2023	VGG-16 model	Fake voice classification dataset	99.0
[*Our*]	2024	**MFCC-GNB XtractNet + GNB**	**Deep fake voice classification dataset**	**99.93**

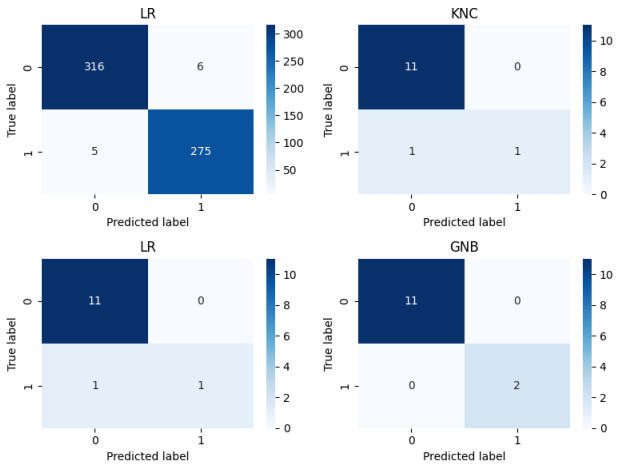

FIGURE 5. Our novel proposed shows a confusion matrix heatmap graph for the detection of deep fake voice.

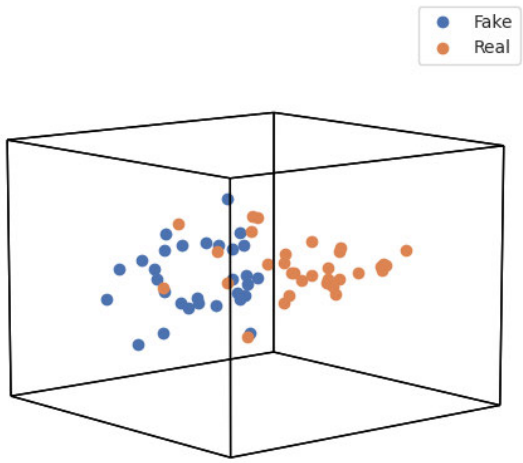

FIGURE 6. Our proposed approach feature space analysis.

G. FEATURE SPACE ANALYSIS

We conducted an in-depth 3D feature space analysis [29] to distinguish between real and fake voices, a critical step in understanding the underlying patterns and relationships within the data. By projecting the features into a three-dimensional space, we are able to visualize how real and fake voice samples are distributed relative to each other. This analysis revealed distinct clusters and boundaries that separate genuine voice data from manipulated or synthesized audio, highlighting the unique characteristics of each category.The figure 6 shows feature space analysis.The 3D visualization not only provided a more intuitive understanding of the feature interactions but also allowed us to identify potential outliers or overlapping regions that could affect model performance. By analyzing these spatial relationships, we could refine our feature extraction and selection processes, ensuring that the most discriminative features were used to distinguish real from fake voices. This detailed 3D feature space analysis

contributed significantly to enhancing the accuracy and robustness of our voice detection model allowing for more precise identification of deepfake audio in complex scenarios.

V. CONCLUSION

this study demonstrates the effectiveness of the MFCC-GNB XtractNet model for deepfake voice detection, achieving an outstanding accuracy score of 99.93%. By extracting MFCC features and transforming them with a Gaussian Naive Bayes model and Non-Negative Factorization, we developed a highly discriminative feature set that significantly enhances detection capabilities. The comprehensive evaluation, including k-fold cross-validation and computational cost analysis, confirmed the robustness and efficiency of our approach. The results highlight the potential of MFCC-GNB XtractNet as a powerful tool for real-world applications, providing reliable and rapid detection of deepfake audio, which is crucial in maintaining the integrity of voice authentication systems.

A. LIMITATIONS

Despite its high accuracy, the MFCC-GNB XtractNet model has several limitations. Its performance may vary with different voice datasets, particularly those with unique accents or recording conditions. The reliance on MFCC features might miss some nuances of advanced deepfake techniques, potentially reducing effectiveness against new manipulation methods. Additionally, the computational cost for feature extraction and transformation could be high, which may impact real-time processing scenarios. Addressing these limitations is essential for enhancing the model's robustness and applicability.

B. FUTURE WORK

Future work will focus on enhancing the MFCC-GNB XtractNet model by exploring additional feature extraction techniques to capture a broader range of voice characteristics. We plan to investigate the model's robustness against emerging deepfake technologies and optimize computational efficiency for real-time applications. Additionally, expanding the model's capability to handle diverse voice datasets and incorporating adversarial training methods will be key to improving its generalizability and resilience.

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